

Benchmark Contamination as a Privacy/Security Vulnerability in Large Language Models (working draft, Pythia-160m)

Large language models are ranked and certified as safe using public benchmarks, and that practice is only valid if the benchmark was absent from pre-training. When a benchmark does appear in the training corpus, a condition known as *benchmark contamination*, the usual concern is that scores overstate capability. A growing line of argument treats contamination instead as a privacy warning sign, on the reasoning that contamination reveals *memorization*, and that memorization is the mechanism by which sensitive training content escapes. That reasoning is rarely stated precisely, and contamination and sensitive-data leakage are distinct phenomena, so we begin by decomposing it into four explicit assumptions and identifying which are empirically testable. Our experiments target the first and most basic of them: that a contamination score reflects retention of a *specific* item rather than how intrinsically predictable that item is. We test it on Pythia models trained on the public Pile, so that training-set membership is ground truth rather than an inferred label, and we compare four existing reference-free detectors (LOSS, Min-K%, Min-K%++, and a compression ratio) against a per-item extraction outcome. We propose no new detector, attack, or metric. Our contribution is a controlled measurement of *incremental* value: a pre-registered partial-correlation and mediation analysis that asks what each detector predicts once the model's per-item loss is held fixed. The three adjusted detectors add no positive predictive value beyond loss. They are near-collinear with it (Spearman 0.74 to 0.90), and their residual associations are null or weakly negative. We do not build on those negative residuals, because their attribution is unstable across near-equivalent specifications: under the pre-registered primary control the only one that is significant after correction belongs to a detector whose collinearity is moderate, so a suppression artifact does not by itself explain it. We therefore claim only the conservative half of the result, the absence of any positive residual, which survives non-linear controls for loss and survives deduplication, and is not explained by token frequency or by the mostly-zero outcome. We read this as a measurement-validity result: the detectors an auditor can afford carry little information about *which* items leak that loss does not already carry, which bounds what detector-based contamination auditing can support. We do not demonstrate leakage of sensitive information. A measurement of personally identifiable information (PII), meaning content that identifies a specific person such as a name paired with an email address or a phone number, on the Enron Emails subset of the Pile returned no detected leakage at our scale, and we report that null rather than treating the sensitive-data link as established. **These results are preliminary, obtained on the smallest (160M) Pythia model on CPU, which is the regime least likely to exhibit memorization. The pipeline is built so that replication at larger scale is a single configuration change.** All analyses are pre-registered and every number is reproducible from a seeded script.

Introduction

Benchmarks, and the assumption they depend on

Large language models (LLMs) are compared, selected, and increasingly certified as safe on the basis of their scores on public test sets called *benchmarks* (Cobbe et al., 2021; Hendrycks et al., 2021). A benchmark is a fixed collection of questions with known answers. A model that answers more of them is taken to be more capable.

That inference depends on one assumption: the model must not have seen the benchmark during training. If it has, a high score may reflect recall of the answer rather than the ability the benchmark was built to measure. The presence of evaluation data inside a model's training corpus is called *benchmark contamination* (Golchin & Surdeanu, 2024).

Contamination is now difficult to avoid. Benchmarks are small, fixed, and copied widely across the web once published. Training corpora are enormous and lightly filtered, assembled at the scale of hundreds of gigabytes to petabytes (Common Crawl Foundation, n.d.; Gao et al., 2020). Benchmark items are therefore drawn into the next crawl through ordinary redistribution, with no adversary and no misconduct required. The research community treats this primarily as a measurement-hygiene problem, on the grounds that a contaminated score overstates capability (Ravaut et al., 2024).

Why this is also a privacy and security question

There is a second consequence that the hygiene framing sets aside. The mechanism that makes a contaminated benchmark score untrustworthy is *memorization*: the model has retained information specific to individual training examples rather than only the general patterns across them. Memorization is also the mechanism behind a known privacy failure. Models trained on web-scale corpora can reproduce training text word for word, including personal information such as names, addresses, and contact details (Carlini et al., 2021; Carlini et al., 2023).

This suggests an appealing argument: because contamination is easy to look for and sensitive-data leakage is not, perhaps a cheap contamination signal can act as an early warning for leakage risk. If so, contaminating a benchmark would not merely distort a leaderboard. It would expose a measurable channel into the training data.

We want to be careful here, because this argument is not automatic, and stating it loosely has consequences for what the resulting measurements mean. Contamination and sensitive-data leakage are related but genuinely distinct phenomena. Contamination concerns *evaluation* data appearing in a training corpus. Leakage concerns a model *emitting* sensitive content. One does not imply the other. Benchmark items are in fact usually public and non-sensitive by construction, so a contaminated benchmark is not itself a privacy harm. Any bridge between the two must be built from explicit premises about a shared underlying mechanism.

Section [2.5](#sec:chain) therefore sets out that bridge as four named assumptions (A1 through A4) rather than as a claim, and states plainly which of them this paper tests, which it inherits from prior work, and which remain untested. The short version is that our experiments bear on the first link, whether a contamination score reflects item-specific retention at all, and that the link from retention to *sensitive* content remains an assumption of the framing rather than a finding of this paper.

The practical problem: which instrument should an auditor trust?

Suppose you want to know whether a model has memorized a particular document. The direct test is to try to make the model reproduce it, which requires generating text and comparing it against the original. That is expensive at corpus scale and requires possessing the document already.

The cheaper alternative, and the one widely used in practice, is a *detector*: a score computed from the model's own reported probabilities for the text, with no retraining and no reference

models. Several such detectors exist, and Section [2.4](#sec:concepts) defines each one before we use it. The simplest is the model’s *loss* on the text, which measures how unsurprised the model is by it. Others (Min-K%, Min-K%++, and a compression-based ratio) are transformations of the same per-token probabilities, designed to separate training members from non-members more sharply than loss alone.

These detectors are developed and ranked by how well they answer a membership question: was this text in the training data? But an auditor concerned with privacy wants the answer to a different question: how much of this specific item has the model retained, and would it come back out? Those two questions need not have the same answer, and the distinction has been noted before. Hayes et al. (2025) report no correlation between a strong membership attack’s success and extraction on the models they study, and suggest the two capture different signals.

Our question is the one that follows from theirs, and it is a question about measurement validity rather than about attack strength. If the detectors an auditor would actually reach for are all computed from the same per-token probabilities as loss, do they carry any information about leakage that loss does not already carry? This matters because it determines whether detector-based auditing can be trusted to tell you *which* items are at risk, or only whether contamination happened at all.

Research questions

- **RQ1 (conceptual validity).** Under what assumptions can a contamination score serve as an indicator of memorization, and further as an indicator of sensitive-information leakage? Which of those assumptions are empirically testable in a setting where training-set membership is known with certainty?
- **RQ2 (incremental predictive value).** Do the widely used detectors provide information about which individual items a model has memorized, over and above what the model’s per-item loss already provides?
- **RQ3 (choice of instrument).** These detectors were designed and tuned to answer one question well, namely whether a text was in the training data. Do the design adjustments *intended* to sharpen that discrimination also help predict whether the model will reproduce the text, or do the two objectives come apart? We note in advance that at our scale no detector separates members from non-members above chance (Section [6.1](#sec:res-membership)), so RQ3 is answered here only in the weaker, design-level sense.

Approach in brief

We study Pythia models (Biderman et al., 2023), which were trained on The Pile (Gao et al., 2020), a corpus that is public. This is the central design choice of the paper and Section [5](#sec:eval) defends it in detail. Because the corpus is published, we can look up whether a document was genuinely in the training data instead of inferring it. Membership is therefore ground truth, which removes the largest source of error in this literature, namely that a detector can appear to succeed because the member and non-member texts differ in topic or era rather than in membership (Duan et al., 2024).

For each document we compute the detector scores, and separately measure a concrete leakage outcome: given the first part of the document, does the model reproduce the rest (Carlini et al., 2023)? We then ask whether the detector scores predict that outcome once the model’s loss is held fixed. Section [2.4](#sec:concepts) explains what "holding loss fixed" means and why it is the right comparison, and Section [5](#sec:eval) explains why we pre-committed to the analysis in writing before running it.

Contributions, and explicit non-contributions

We are deliberate about what this paper is not. It introduces no new detector, no new attack, and no new metric. Every detection method we run is from prior work (Brown et al., 2020; Carlini et al., 2021; Oren et al., 2024; Shi et al., 2024; Yeom et al., 2018; J. Zhang et al., 2025). We do not train or fine-tune models, and we do not attack deployed systems for real third-party personal data. Within that scope:

- **An explicit conceptual bridge, stated as assumptions.** We replace the usual one-line assertion that contamination indicates leakage risk with four named assumptions, and we identify which are testable, which we test, and which the field currently takes on faith (Section [2.5](#sec:chain)). This makes the framing falsifiable and bounds what any contamination-based privacy claim can support.
- **A ground-truth comparison of existing detectors as leakage predictors.** On Pythia and The Pile, where membership is known, we evaluate the detectors both as membership classifiers and as predictors of a per-item extraction outcome, using the low-false-positive reporting convention that privacy work expects (Carlini et al., 2022) and controlling for the frequency, duplication, and distribution-shift confounds that prior work identifies (Biderman et al., 2023; Duan et al., 2024).
- **A controlled measurement of incremental value, which is the paper’s core result.** Using an analysis fixed in advance, we ask whether each detector predicts leakage after the model’s per-item loss is accounted for. In our setting the detectors add no positive predictive value beyond loss. They are close to being mathematical restatements of loss, which we quantify directly, and we therefore read the result as a measurement-validity finding: these instruments carry little information about *which* items leak that loss does not already carry.

What we do not claim.

We do not claim to demonstrate leakage of sensitive or personal information. We attempted such a measurement on the Enron Emails subset that sits inside The Pile, and it returned no detected leakage at the scale we could run (Section [6](#sec:results)). We report that null honestly and treat the step from memorization to *sensitive* content as an untested assumption (A3 and A4 in Section [2.5](#sec:chain)) rather than as a result. We also do not claim a stronger attack than prior work, and we do not claim that the detectors are *worse* than loss, only that we find no evidence they are better.

Preliminary scope.

The results reported here come from the smallest Pythia model, at 160 million parameters, run on a CPU. Memorization increases with model scale (Carlini et al., 2023), so this is the regime least likely to show leakage, and we flag every affected conclusion as preliminary. The pipeline was built so that repeating everything on larger models is a change of one configuration value.

Roadmap

Section [2](#sec:background) explains benchmarks, memorization, and the technical terms and statistical tools this paper relies on, and then states the assumption chain that connects contamination to leakage. Section [3](#sec:threat) defines the threat model. Section [4](#sec:relatedwork) positions the work against prior contamination, memorization, and membership-inference research. Section [5](#sec:eval) gives the experimental design and the reasoning behind each methodological choice. Section [6](#sec:results) reports results, Section [7](#sec:discussion) interprets them, and Section [8](#sec:limitations) states the limitations.

Background and Key Concepts

Benchmarks as proxies for capability

LLM benchmarks act as *proxies* for abilities that cannot be measured directly, such as reasoning, comprehension, factual recall, and programming skill. By scoring a model on a fixed set of standardized tasks, the community infers how useful, and increasingly how safe, that model is likely to be in deployment (Hendrycks et al., 2021). Familiar examples target distinct competencies: MMLU covers multitask knowledge across 57 subjects (Hendrycks et al., 2021), GSM8K covers multi-step arithmetic word problems (Cobbe et al., 2021), and HumanEval covers functional code generation (M. Chen et al., 2021). Scores on these suites drive model selection, leaderboard position, and public claims of progress.

The validity assumption, and what breaks when it fails

The inference from benchmark score to capability rests on one strict assumption: *the test data was not seen during training*. Only then does a high score support the intended conclusion, that the model *generalizes*, meaning it applies learned regularities to inputs it has not encountered, rather than *memorizes*, meaning it reproduces specific training instances.

When the assumption fails, the benchmark stops measuring what it claims to measure. A memorized test item raises the score with no corresponding gain in ability, so the metric becomes an unreliable estimate of the underlying construct. This distinction is not only conceptual. Memorization is directly measurable as verbatim regeneration of training text, and it grows predictably with model size, with how often a sequence was duplicated in training, and with how much context the model is given (Carlini et al., 2023). A closely related privacy-facing measurement plants a secret in the training data and records how strongly the model prefers it over random alternatives, which rises with the number of times the secret was seen (Carlini et al., 2019). Memorization is also highly uneven across examples rather than a uniform background rate, so which particular items a model memorizes is itself a measurable

per-example property (C. Zhang et al., 2023).

Static test sets meet weakly filtered corpora

The structural tension is that benchmarks and training corpora have opposite properties. Benchmarks are static, small, widely circulated, and publicly indexed. Once an MMLU or GSM8K item is published it is copied into papers, blog posts, code repositories, and forum discussions. Training corpora are the reverse: very large web scrapes with light filtering. Common Crawl (Common Crawl Foundation, n.d.) and The Pile (Gao et al., 2020) are assembled at the scale of hundreds of gigabytes to petabytes, where reliably removing any particular short string is impractical.

The consequence is that benchmark items enter training corpora through ordinary redistribution, with no adversary required. A public benchmark is therefore a persistent and low-effort exposure surface, because the very properties that make a benchmark useful (stable, shared, citable) are what make its eventual presence in a future crawl likely.

Key concepts and methodological tools

This subsection defines every technical idea used later in the paper. Each entry states what the concept is, why the study needs it, and how it bears on the research questions of Section [1](#sec:intro).

Token probabilities, loss, and perplexity

A language model reads text as a sequence of *tokens*, which are words or word fragments. At each position it produces a probability distribution over which token comes next. Given a real text, we can therefore ask what probability the model assigned to each token that actually occurred.

The *loss* of a text is the average negative logarithm of those probabilities, so a low loss means the model found the text unsurprising. *Perplexity* is the exponential of the loss and carries the same information on a different scale, so we use loss throughout. Every detector in this paper is computed from these same per-token probabilities, which is the fact that ultimately drives our result.

Orientation convention. All four detector scores in this paper are oriented so that **higher means more member-like**. For the loss-based score this means we report the mean per-token log-probability, which is the negation of the loss as defined above, and which is therefore higher for text the model finds unsurprising. We state this explicitly because the sign is easy to invert when reading the tables, where a positive correlation between the loss-based score and extraction means *less surprising text is more extractable*.

Why this is needed. Loss is the oldest and simplest membership signal, on the reasoning that a model tends to find its own training data less surprising than new text (Yeom et al., 2018). It is also the natural baseline against which any more elaborate detector must prove its worth, which makes it the control variable in RQ2.

Memorization and how extraction is measured

We use the standard operational definition of *extractable memorization* (Carlini et al., 2023). Take a document from the training data, split it into a *prefix* and a *suffix*, and give the model only the prefix. Then let the model continue the text by repeatedly choosing its single most probable next token, a procedure called *greedy decoding*. If the continuation reproduces the suffix, the suffix is *extractable*: the model has retained enough about this specific document to

regenerate it.

We report two versions of this outcome. *Exact extraction* is whether the whole suffix was reproduced. *Fractional extraction* is the proportion of leading suffix tokens reproduced before the first mismatch, which is a softer measure and is much less prone to being zero for every item.

The *extraction rate* is the fraction of sampled documents that were exactly extractable.

Throughout this paper we use *extraction* for the outcome we actually measure, namely verbatim regeneration of a held-out suffix whatever its content, and we reserve *leakage*, or *sensitive-data leakage*, for the emission of *sensitive* content, which we do not measure. Assumption A3 in Section [2.5](#sec:chain) is precisely the gap between the two, and keeping the words separate is what stops that gap from being crossed by accident.

Why this is needed. Extraction is our concrete outcome, and the quantity detectors must predict if they are to be useful for privacy auditing. It is a lower bound rather than a complete account, because a determined adversary may use richer prompting than a single prefix (Nasr et al., 2025), and because rewording can preserve content while defeating exact matching (Ippolito et al., 2023).

Personally identifiable information

Personally identifiable information (PII) is content that identifies a specific person, such as a name paired with an email address, a phone number, or a postal address. Leakage of PII from training data is the canonical privacy harm in this literature (Carlini et al., 2021), it decomposes into distinct extraction, reconstruction, and inference threats (Lukas et al., 2023), and models leak it more through memorization than through inference about individuals (Huang et al., 2022).

Why this is needed. PII is the harm that motivates the security framing. It is also the point where we must be most careful about scope, because benchmark items are generally not PII-bearing. Section [2.5](#sec:chain) makes this gap explicit.

Membership inference, and the detectors we evaluate

Membership inference asks whether a particular piece of text was part of a model’s training data (Shokri et al., 2017). A *detector* assigns each text a score intended to be higher for training members.

Detectors divide by what they require. *Reference-based* methods compare the target model against other models trained on similar data, which is accurate but requires training those extra models and is therefore infeasible at the scale of a corpus like The Pile (Carlini et al., 2022). *Reference-free* methods need only the target model’s own token probabilities. Reference-free methods are what a real auditor can afford, so they are the family we study. We evaluate four:

- **LOSS.** The average per-token loss described above, used directly as the score (Yeom et al., 2018). This is the baseline.
- **Min-K%.** Rather than averaging over all tokens, average only over the $k\%$ of tokens the model found *least* likely. The motivation is that a text the model has seen should lack highly surprising outlier tokens, so focusing on the worst tokens should sharpen the distinction (Shi et al., 2024). We use $k = 20$.

- **Min-K%++**. A refinement that, before selecting the least likely tokens, rescales each token's log-probability by the mean and standard deviation of the model's full distribution at that position. This asks whether a token was unlikely *relative to the alternatives the model considered there*, rather than in absolute terms (J. Zhang et al., 2025). It needs the full next-token distribution and therefore assumes deeper access than the others.
- **zlib ratio**. Divide the model's loss by the length of the text after generic compression with the zlib algorithm. Compressed length is a model-independent estimate of how repetitive or formulaic a text is, so the ratio is intended to discount texts that are unsurprising merely because they are boilerplate (Carlini et al., 2021).

Throughout, we call the last three *calibrated* detectors, meaning only that each adjusts raw loss by some additional quantity in an attempt to improve separation. The word carries no claim that the adjustment is probabilistically well calibrated.

Note that loss appears in two roles in this paper. As **LOSS** it is one of the four detectors, and the baseline one. As **loss** it is also the control variable in RQ2, the quantity we hold fixed to ask whether the other three add anything. This dual role is deliberate, because the question of whether a detector adds value only has meaning relative to the cheapest available alternative. It does mean that RQ2 is a question about the three *calibrated* detectors and not about LOSS itself.

When a membership detector is applied to a benchmark item, its score is conventionally read as a *contamination score*. We use the two terms for the same number, and we flag the assumption this involves as A0 in Section [2.5](#sec:chain).

Why this is needed. These four are the instruments an auditor would actually use, and RQ2 asks whether the three calibrated ones add anything to the plain baseline.

Levels of access to a model

Detectors differ in what they need from the model. *Black-box* access means text in and text out, with no numeric scores, which is what a public chat interface provides. *Gray-box* access adds the model's log-probability for each token of a text you supply. *White-box* access adds the model's full probability distribution over the entire vocabulary at every position.

Why this is needed. Section [3](#sec:threat) grades each detector by the least access it requires, and this determines who can actually run it. An external auditor of a hosted model typically has black-box access only. Of our four detectors, LOSS, Min-K%, and the zlib ratio need gray-box access, and Min-K%++ is the only one that requires white-box access.

Corpus-side and benchmark-level contamination tests

Two further tests operate on the corpus or the whole benchmark rather than on the model's probabilities for a single item.

n-gram overlap searches the training corpus for exact matches of contiguous n -token spans from a benchmark item, the convention introduced with GPT-3 (Brown et al., 2020). Because it needs the corpus rather than the model, we use it to build contamination labels rather than as an attack. It provides a lower bound only, since reworded copies do not match.

The *permutation*, or *exchangeability*, test asks whether a model prefers a benchmark's items in their original published order over randomly shuffled orderings. A model trained on the benchmark file tends to favour the canonical order, which yields a statistically calibrated

benchmark-level test (Oren et al., 2024).

Why this is needed. These give benchmark-level and corpus-level evidence to complement the per-item model-side detectors, and they let us report contamination status for real benchmarks independent of any detector’s reliability.

How privacy attacks should be evaluated

A detector produces a score, and turning it into a decision requires a threshold. Varying the threshold traces a *receiver operating characteristic* (ROC) curve, plotting the *true-positive rate* (the fraction of real members flagged) against the *false-positive rate* (the fraction of non-members wrongly flagged). The *area under the curve* (AUC) summarizes the whole curve in one number, where 0.5 means no better than chance.

Average-case AUC is a poor summary for a privacy threat, because an attack matters if it identifies *some* members with very few false accusations. The accepted convention is therefore to report the true-positive rate at a low fixed false-positive rate, such as 0.1% or 1%, and to plot the ROC curve on logarithmic axes so that the low-false-positive region is visible (Carlini et al., 2022). We follow that convention and treat AUC as secondary.

Why this is needed. It is the reporting standard that privacy and security venues expect, and it prevents a detector that never fires confidently from looking successful.

Statistical tools for isolating incremental value

RQ2 asks whether a detector adds predictive value *beyond* loss, which is a question about incremental contribution rather than about raw association. The following tools answer it, and each is used exactly as defined here.

- **Spearman rank correlation (ρ).** A measure between -1 and $+1$ of how well two quantities increase together after each is replaced by its rank. Ranks make it insensitive to outliers and to any monotone rescaling, which matters because the detectors are on different and arbitrary scales.
- **Partial correlation.** The correlation between a detector score and the leakage outcome *after statistically removing* the part of each that is predictable from loss. This is the precise meaning of "holding loss fixed". If a detector’s partial correlation is zero, it tells us nothing about leakage that loss had not already told us. This single quantity is the paper’s primary outcome measure.
- **Collinearity.** Two variables are collinear when one is close to a rescaling of the other. We measure it with the correlation between loss and each detector, and with the *variance inflation factor*, a standard index of how much collinearity destabilizes a regression coefficient. Collinearity matters because it limits how confidently any residual effect can be interpreted.
- **Suppression.** When two predictors are strongly collinear, the estimated coefficient of one can flip sign for purely algebraic reasons rather than because the relationship truly reverses. This is a known artifact, and it is why we interpret a negative partial correlation cautiously and claim only the absence of positive value.
- **Mediation.** A descriptive decomposition of an association into the part that travels through an intermediate variable (here loss) and the part that does not. We use it only as description, not as a causal claim, because our design is observational.

- **Non-linear controls.** Partial correlation removes only a linear relationship with loss. To check that nothing survives merely because the true relationship is curved, we repeat the analysis after removing a cubic function of loss, and again within narrow bands of similar loss values.
- **Bootstrap confidence interval.** Repeatedly resample the items with replacement, recompute the statistic, and report the middle 95% of the results. This conveys how much the estimate would move with a different sample of the same size.
- **Permutation test.** Randomly shuffle one variable many times to build the distribution of the statistic under no association, then locate the observed value in it. This yields a p -value without assuming the data are normally distributed.
- **Multiple-comparison correction.** Testing several detectors at once inflates the chance of at least one false positive. We apply the Benjamini-Hochberg procedure, which controls the expected proportion of false discoveries among the findings declared significant.
- **Zero inflation.** An outcome is zero-inflated when most observations are exactly zero, which weakens any correlation and widens its confidence interval. Our exact-extraction outcome is severely zero-inflated at small model scale, which is why we also report the fractional outcome.
- **Pre-registration.** Writing the analysis plan down, in the repository, before running it. This prevents the outcome from being chosen after seeing the data, which is the main way a null result can be quietly converted into a positive one.
- **Zero-order correlation.** The plain correlation between two quantities with nothing controlled for. We use it as the contrast to the partial correlation, so that the difference between the two isolates what loss accounts for.
- **Decile stratification.** The coarser version of the non-linear control described above. Sort the items by loss, cut them into ten equal groups, compute the correlation inside each group, and combine. It removes any relationship with loss, whatever its shape, at the cost of statistical power.
- **Kendall's τ -b.** An alternative rank correlation that handles ties explicitly. We report it as a robustness check because our outcome has many tied values at zero, which Spearman handles less gracefully.
- **Deduplication.** Removing near-duplicate documents from a training corpus before training. Duplication is one of the strongest known drivers of memorization (Carlini et al., 2023), so the Pythia suite ships a deduplicated-corpus counterpart to each model, which we use as a robustness arm.
- **Distribution shift, and the temporal confound.** A detector can appear to identify training members when the member and non-member texts simply differ in some other way, such as topic, source, or date of writing. The most common instance is the *temporal confound*, where non-members are drawn from a later period than the training cutoff, so the detector is really detecting era rather than membership (Duan et al., 2024). Avoiding this is the main reason we construct members and non-members from the same corpus and match them by subset.
- **Construct validity.** Whether a measurement actually captures the concept it is meant to capture, as opposed to something correlated with it. This is the central methodological risk in our design, because our outcome is computed from the same token probabilities as our

predictor, and Section [8](#sec:limitations) treats it directly.

- **Differential privacy.** A formal guarantee, added during training, that the model's behaviour changes only slightly when any single training record is removed, which bounds how much can be inferred about that record ([abadi2016dp](#)). We discuss it as the producer-side mitigation our threat model motivates and do not implement it.

For the variance inflation factor, a value of 1 means no collinearity, values above roughly 5 are conventionally treated as high, and our detectors range from 2.4 to 6.2. We report the index rather than applying a threshold mechanically.

From contamination to leakage: the assumption chain

The security framing of Section [1](#sec:intro) rests on a chain of reasoning from contamination to sensitive-information leakage. Contamination and leakage are distinct phenomena, and the chain is not automatic, so we state it as four assumptions and then say which the paper tests. Writing them out serves two purposes: it makes the framing falsifiable, and it bounds what any contamination-based privacy claim is entitled to conclude.

The chain has the form contamination → memorization → extractable output → sensitive leakage, and each arrow needs a premise.

- **A0 (metric identification).** The scores used to detect contamination are the same scores used to detect training-set membership, and they behave on benchmark items as they do on other training documents. *Why it is needed.* Contamination detection in practice reuses the membership-inference toolkit unchanged, in that a benchmark item is called contaminated when a membership detector scores it as a likely training member. Every quantity we compute is therefore a membership score, and calling it a contamination score presupposes that the two coincide. *Status.* **Assumed, not tested.** Our per-item analysis is run on Pile documents rather than on benchmark items, because that is where membership is ground truth and where an extraction outcome can be measured. Benchmark items are handled separately by the corpus-side and benchmark-level tests. Benchmark items are short, highly templated, and unusually widely duplicated, all of which are known to affect memorization (Carlini et al., 2023), so this identification is not free and we flag it as the chain's entry condition.
- **A1 (signal validity).** A detector's contamination score reflects *item-specific retention*, and not merely how intrinsically predictable the text is. *Why it is needed.* If a score rises only because a text is generic or formulaic, it cannot separate "the model saw this item" from "this item is easy for any model", and it cannot indicate which items carry risk. *Status.* Testable wherever membership is ground truth, and **this is the assumption our experiments examine, for the three calibrated detectors.** RQ2 is its operational form. Two caveats bound what we can conclude, and we state them here rather than in the limitations because they shape the claim itself. First, we hold intrinsic predictability fixed using loss, but loss is itself the simplest retention signal, so a null residual establishes only that a calibrated detector adds nothing *beyond the cheapest available retention signal*, not that it carries no retention signal at all. We cannot test A1 for LOSS itself, because loss is the control. Second, at our scale loss does not separate members from non-members above chance (Section [6.1](#sec:res-membership)), and part of the association between loss

and extraction is definitional, since both are computed from the same per-token probabilities. Read together, our evidence is consistent with A1 failing at this scale for every detector we test, including the baseline, and we regard establishing A1 for *any* affordable detector as an open problem rather than a settled premise.

- **A2 (retention implies emission).** Content the model has retained can be elicited by a procedure an adversary could actually run. *Why it is needed.* Retained but unreachable content is not a leak. A privacy claim requires that something come out. *Status.* Instantiated here rather than tested. We implement one realizable elicitation procedure, prefix-continuation with greedy decoding (Carlini et al., 2023), and observe a low but non-zero rate at our scale. It is a lower bound, since stronger prompting extracts more (Nasr et al., 2025). Note also that this procedure requires the adversary to already hold a prefix of the target document, so it models *targeted* extraction against a known document rather than untargeted discovery, which is a weaker threat than the phrase "an adversary could run it" might suggest.
- **A3 (sensitivity co-occurrence).** The content a model memorizes includes sensitive information. *Why it is needed.* This is the assumption that turns an evaluation problem into a privacy problem, and it is the weakest link. Memorizing a public MMLU question damages measurement validity but harms nobody's privacy. Benchmark items are generally public and non-sensitive by construction. *Status.* Testable in principle on a corpus with known sensitive content. We attempted it on the Enron Emails subset of The Pile and detected no leakage at our scale (Section [6](#sec:results)), so for practical purposes it remains **untested here**. Prior work establishes that models can emit PII from web-scale corpora (Carlini et al., 2021; Lukas et al., 2023), which is why we regard the assumption as plausible rather than established.
- **A4 (mechanism transfer).** The relationship measured on benchmark-like documents also holds for sensitive records. *Why it is needed.* We measure on Pile documents, whereas the privacy claim concerns sensitive content. If memorization behaves differently for rare, unusually formatted, or seldom-duplicated strings, then a result about benchmark-like text does not automatically transfer. *Status.* **Untested here**, and non-trivial. Memorization is known to be strongly example-dependent (C. Zhang et al., 2023) and to scale with duplication (Carlini et al., 2023), both of which argue against assuming free transfer.

What this means for the paper's claims.

Our empirical work bears on A1, and secondarily on A2. We do not establish A3 or A4, so we make no claim to have demonstrated sensitive-data leakage, and we present the privacy framing as the motivation for studying A1 rather than as a validated end-to-end result. This matters for interpretation in a specific way. A1 is the chain's first plank, so evidence that it fails for the calibrated detectors constrains every downstream privacy claim built on those detectors, regardless of whether A3 and A4 hold. Conversely, and we state this plainly, a positive result on A1 alone would not have demonstrated a privacy harm either. Section [8](#sec:limitations) returns to what would be required to test A3 and A4 directly.

Threat Model

This section states who is acting, what they know, what they are trying to achieve, and what counts as success. All technical terms used here are defined in Section [9](#)

[2.4](#sec:concepts).

Two parties with different problems

Discussions of contamination often blur two roles that need separating, because they have different access and different objectives.

The **auditor** is a defender. This is a model developer, an evaluation organization, or an external reviewer who wants to know whether a benchmark leaked into training and, more usefully, *which* specific items a model has retained. The auditor is willing to run cheap tests over many items and cannot afford to train reference models. The auditor is the party this paper is really about, because the question of whether a detector is a valid instrument is the auditor's question.

The **adversary** is an attacker who wants to recover content from the training data. The adversary needs only the deployed model and does not need the training corpus.

A key asymmetry is that the auditor is trying to *predict* what the adversary could extract, without performing the extraction on every item. That is precisely why detector validity matters, and it is the practical stake behind assumption A1 in Section [2.5](#sec:chain).

Goals, in increasing order of harm

- **G1: membership inference.** Decide whether one specific sequence, such as a benchmark item, document, or record, was in the training corpus.
- **G2: benchmark-level contamination confirmation.** Decide, with a controlled false-positive rate, whether an entire benchmark was trained on.
- **G3: extraction.** Cause the model to reproduce content that was in the training data. On a corpus with known sensitive content this is where privacy harm would occur.

G3 is the outcome that matters. G1 and G2 are of interest mainly insofar as they predict G3, and whether they do is exactly what this paper measures. Under the terminology of Section [2.5](#sec:chain), a detector that achieves G1 but fails to predict G3 is a detector for which A1 does not hold in the sense the privacy framing requires.

Knowledge and access

We grade the detectors by the least access each one needs, using the definitions from Section [2.4](#sec:concepts):

- **Black-box.** Text in, text out, with no probabilities. Guided prompting falls here. We do not evaluate black-box detectors.
- **Gray-box.** The model's per-token log-probabilities for a supplied text. LOSS, Min-K%, and the zlib ratio need only this.
- **White-box.** The full next-token probability distribution at each position. Min-K%++ requires this, because it rescales each token against the alternatives the model considered.

The n -gram overlap test is different in kind, because it inspects the training corpus rather than the model. We can run it only because The Pile is public, and we use it to construct

contamination labels rather than as an attack an adversary could mount. Extraction (G3) requires only the ability to prompt the model and read its output.

For our experiments the auditor additionally knows the training corpus, which is what makes membership ground truth rather than an inference. We treat this as a deliberate methodological advantage rather than a realistic assumption, and Section [5](#sec:eval) explains why studying a model whose corpus is public is the only way to test A1 without confounding.

Success criteria

- **G1.** True-positive rate at 0.1% and at 1% false-positive rate, read from a logarithmic-axis ROC curve, with AUC reported as secondary and with bootstrap confidence intervals. Section [2.4](#sec:concepts) explains why the low-false-positive operating point is the meaningful one for a privacy attack.
- **G2.** A permutation-test p -value below a pre-specified threshold, with a controlled false-positive rate (Oren et al., 2024).
- **G3.** A non-zero extraction rate, and then the paper’s central criterion: a *positive* association between a per-item detector score and the per-item extraction outcome that *survives holding the model’s loss fixed*, in the partial-correlation sense of Section [2.4](#sec:concepts).

The last criterion is the one that distinguishes a detector that genuinely identifies which items are at risk from one that merely restates the model’s loss in different units. We fixed this criterion in advance of running the analysis, for the reason given under pre-registration in Section [2.4](#sec:concepts).

Out of scope

We do not attack deployed production models for real third-party personal data. We do not train or fine-tune models, and we propose no new detector. We do not claim to demonstrate leakage of sensitive information, since assumptions A3 and A4 of Section [2.5](#sec:chain) are untested here. Differential privacy is discussed as the producer-side mitigation this threat model motivates (Section [4.5](#sec:dp)) and is not implemented or evaluated.

Related Work: Contamination, Memorization, and Privacy Leakage

Defining benchmark contamination

We adopt the standard definition: *benchmark contamination* is the presence of evaluation data, inputs, labels, or accompanying metadata, within a model’s pre-training corpus (Golchin & Surdeanu, 2024). Contamination matters for two reasons. First, it invalidates evaluation: a contaminated score conflates capability with retrieval, so the metric no longer estimates generalization. Second, contamination is often argued to be a measurable proxy for unintended memorization, and memorization is the mechanism behind sensitive-content

leakage. That argument is not automatic, and Section [2.5](#sec:chain) decomposes it into four assumptions A1 through A4. Our empirical object is the first link, A1, and not the chain as a whole.

A typology of contamination

Following the project’s framing and the contamination-detection survey (Ravaut et al., 2024), we distinguish three forms by the transformation between the corpus copy and the benchmark item:

- **Verbatim contamination.** The exact token sequence of a test item appears in training data. This is what classical n -gram decontamination targets (e.g., the 13-gram overlap test introduced for GPT-3 (Brown et al., 2020)) and what verbatim-extraction memorization measures (Carlini et al., 2023).
- **Paraphrased contamination.** The semantic content is present but reworded, so surface-level n -gram matching misses it. A perfect verbatim filter provides only a false sense of safety, since style-transfer rephrasings evade it while preserving the leaked information (Ippolito et al., 2023).
- **Semantic contamination.** The underlying knowledge or answer is encoded without lexical overlap (e.g., the same question-answer mapping in a different format). Detecting it requires model-behavioral or distributional signals rather than string matching.

A second, orthogonal severity axis is *what* is contaminated: input-only leakage inflates familiarity, whereas joint input and label leakage enables direct answer retrieval and is the most damaging to evaluation validity. Empirically, overlap between open-model training data and benchmarks such as GSM8K has been reported for models trained on largely undisclosed corpora (Touvron et al., 2023), motivating ground-truth-controlled study on models whose corpus is fully public.

Why memorization is a security and privacy problem

Memorization is not a benign curiosity. Over-parameterized models trained on web-scale scrapes retain and can regurgitate verbatim sequences, including personally identifiable information (PII) (Carlini et al., 2021). This has been formalized along several axes that we reuse as outcome variables:

- **k -eidetic / extractable memorization.** A string is extractable if a prefix makes the model regenerate it, and is k -eidetic if it occurs in at most k training documents (Carlini et al., 2021). The prefix-continuation form under greedy decoding makes this directly measurable (Carlini et al., 2023).
- **Exposure and example-level memorization.** Injecting a canary secret and measuring its *exposure*, its rank against random alternatives, quantifies unintended memorization and its growth with occurrence count (Carlini et al., 2019). This requires control over the training process (canary insertion), which our pretrained-checkpoint setting does not afford, so we use it for definitions rather than as a measurement. Relatedly, memorization is concentrated on specific examples (C. Zhang et al., 2023) rather than spread uniformly, which is what makes per-item leakage a meaningful target for per-item prediction in the first place. Whether the available per-item scores actually supply that prediction is the question of RQ2, and Section [2.5](#sec:chain) states it as assumption A1.

- **Extraction at scale.** Production models can be driven, via a divergence attack, to emit memorized training data well above their nominal aligned rate, recovering thousands of verbatim examples cheaply (Nasr et al., 2025).
- **PII leakage games.** Leakage of personally identifiable information decomposes into extraction, reconstruction, and inference. Data scrubbing and differential privacy reduce but do not eliminate it (Lukas et al., 2023), models leak PII through memorization more than through associative inference (Huang et al., 2022), and black-box probing tools can elicit a data subject’s PII directly from a deployed model (Kim et al., 2023).

This is where the security framing enters, and it is also where it must be stated carefully. The inference from contamination to privacy vulnerability requires that memorized content include sensitive material (A3) and that the measured relationship transfer to sensitive records (A4). Section [2.5](#sec:chain) states both as assumptions that this paper does not test.

The membership-inference lineage

Deciding whether a specific record was in a model’s training set is the canonical privacy attack, and contamination detection is an instance of it. The lineage we build on runs as follows. *Shadow-model* attacks established the threat: by training reference models on data drawn from the same distribution, an adversary learns to distinguish members from non-members from the target model’s outputs (Shokri et al., 2017). Yeom et al. (2018) tied attack success to overfitting and gave the simplest practical baseline (thresholding the per-example loss) together with the *membership advantage* (TPR–FPR) figure of merit. Carlini et al. (2022)’s *Likelihood Ratio Attack* (LiRA) then reframed MIA from first principles as a per-example hypothesis test calibrated with shadow models, and, central to our methodology, argued that average-case AUC is the wrong yardstick for a privacy threat: an attack matters if it identifies *some* members with very few false accusations, so the right report is TPR at a low, fixed FPR on a log-scale ROC curve. Shadow-model calibration, however, is infeasible at Pile/Pythia scale (it requires training many models on the training distribution), so we adopt LiRA’s *metric* but not its *attack*.

For pre-trained LLMs, the field moved to *reference-free* likelihood signals that need no shadow models. Min-K% Prob averages the log-probabilities of a sequence’s lowest-probability *k*% of tokens, on the hypothesis that members lack high-surprise outlier tokens (Shi et al., 2024). Min-K%++ sharpens this by z-scoring each token against the *full* next-token distribution before averaging, detecting that the target token sits at a local maximum of the modeled distribution (J. Zhang et al., 2025). A parallel reference-free line, neighbourhood comparison, calibrates a sample’s score against synthetically generated neighbour texts instead of a reference model (Mattern et al., 2023). We treat it as a related approach we do not evaluate, since it needs many extra masked-LM forward passes per example and, in the regime below, underperforms. The reality check on this whole line is the MIMIR study: a large-scale audit on Pythia (160M–12B) and The Pile with controlled member/non-member splits finds that these attacks barely exceed chance (AUC $\approx$ 0.5–0.6), that LLMs see their corpus for too few epochs over too large a dataset to memorize in the way classical MIA assumes, and that apparent successes frequently reflect a temporal or topical *distribution shift* between the splits rather than membership (Duan et al., 2024). This finding defines our honesty constraint: we do not claim to beat these numbers. We ask whether the weak signal that remains still predicts leakage.

Differential privacy as the defense direction

The standard principled mitigation for training-data leakage is differential privacy. DP-SGD bounds any single example’s influence on the trained model by clipping per-example gradients and adding calibrated noise, with privacy accounted via the moments accountant (Abadi et al., 2016). Applied to language models, DP fine-tuning can retain much of the utility of non-private training, particularly with large pre-trained backbones (Li et al., 2022) and parameter-efficient adaptation (Yu et al., 2022). DP bounds memorization and thereby the leakage we measure, but at a privacy and utility cost and, crucially for us, it must be applied *at training time*. It is a defense for model producers, not a detector available to an auditor of an already-released model. We therefore position DP as the mitigation our threat model motivates, and do not implement it (we train no models).

Existing detection techniques

We describe the techniques we implement and compare. The comparative evaluation and the access requirements appear in Section [5](#sec:eval). All operate without any novel detector of our own, our contribution is their security-framed, ground-truth evaluation, not a new method.

- ***n*-gram / substring overlap.** Flag a benchmark item that shares an *N*-gram with the corpus (Brown et al., 2020). Requires corpus access, and misses paraphrased and semantic contamination.
- **Loss / perplexity thresholding.** The mandatory membership-inference baseline: members exhibit lower loss, with attack success tied to overfitting (Yeom et al., 2018).
- **Min-K% Prob.** Average the log-probabilities of the lowest-probability *k*% of tokens, reference-free and logprob-only (Shi et al., 2024).
- **Min-K%+ +.** Normalizes each token’s log-probability against the full next-token distribution before the bottom-*k*% average, the most recent refinement in the reference-free line (J. Zhang et al., 2025).
- **zlib ratio.** Calibrate model perplexity by the zlib-compressed size of the text, controlling for intrinsic compressibility/frequency (Carlini et al., 2021).
- **Permutation / exchangeability test.** At the *benchmark* level rather than per item, score each ordering of a benchmark’s examples by the log-likelihood of their concatenation and compare the canonical (published) order against random shufflings. A model trained on the benchmark in canonical order favours it beyond chance, yielding a provable, FPR-controlled contamination certificate (Oren et al., 2024).

We additionally note two techniques we describe but *do not* evaluate, since our ground-truth, logit-access setting makes likelihood-based detectors stronger and cleaner: *guided prompting*, which prompts a model with dataset metadata and a partial instance and tests for verbatim completion (Golchin & Surdeanu, 2024), a black-box signal aimed at closed models, and the reference-free *neighbourhood* and shadow-model *reference* attacks discussed in Section [4.4](#sec:mia-lineage) (Mattern et al., 2023; Shokri et al., 2017).

Limitations of existing detection, and our positioning

Two limitations frame our contribution. First, *detection is fragile to the transformation*: string-matching misses paraphrased and semantic contamination (Ippolito et al., 2023), and

likelihood-based membership inference is known to barely exceed chance on pre-trained LLMs evaluated under controlled ground truth, because the corpora are seen for few epochs and member/non-member boundaries are fuzzy (Duan et al., 2024). Second, *evaluation conventions matter*: average-case AUC or accuracy can mask whether an attack confidently identifies any members, so the security-appropriate report is true-positive rate at low false-positive rate with log-scale ROC (Carlini et al., 2022). We therefore do not claim a stronger detector. We ask a different, security-relevant question: *even where contamination signal is weak, does it predict concrete extraction?* We answer it with ground-truth membership on the Pythia suite (Biderman et al., 2023) trained on the public Pile (Gao et al., 2020), under the low-FPR protocol, with explicit controls for the frequency, duplication, and temporal confounds that prior work identifies.

Closest prior work, and how we differ

Three recent works reach conclusions adjacent to ours, and we are careful to position against them rather than overclaim. The closest prior result is Hayes et al. (2025), who established the divergence we build on: they “observe no correlation with MIA success” for extraction and conclude the “two privacy attacks may capture different signals.” We do not claim that observation. Their evidence is a *direct, zero-order* correlation between a reference-model attack (LiRA) and extraction, and our question is what remains once loss is controlled. We *partial out per-item loss* rather than correlating directly, and we target the reference-free *calibrated* detectors (Min-K%, Min-K%++, zlib) that the contamination-detection literature actually deploys, which converts their observation into a measurement-validity statement about those instruments. Al Sahili et al. (2025) reach a compatible conclusion for targeted extraction, that “complex MIA techniques yield only marginal improvements over simple likelihood-based ranking,” but they establish it through aggregate *ranking-precision* comparisons and an AdaBoost ensemble over MIA features, reporting *marginal gains* rather than testing for independent signal. In contrast, we run a pre-registered *partial correlation controlling for raw per-item loss*, which lets us state a sharper, conservative claim: the reference-free detectors contribute no *positive* residual predictive value once loss is partialled out, with the negative residuals we do observe attributable to suppression under near-collinearity rather than to inverse prediction. Independently, B. Chen et al. (2025) find for the *membership* task that the few detectors numerically above the loss baseline (Min-K%, Min-K%++, ReCaLL) do not beat it robustly once random-seed variance is accounted for, and that performance is domain-dependent (code-like, low-token-diversity domains such as GitHub and StackExchange behave differently from Wikipedia and FreeLaw). We revisit this domain dependence for the *extraction* outcome in our per-domain analysis (Section [6.2](#sec:res-headline)), noting it is a distinct axis from their membership-AUC result. Finally, blind-baseline and SoK critiques (Das et al., 2024; Meeus et al., 2025) show that post-hoc member/non-member splits can make detector “success” an artifact of distribution shift. Our use of ground-truth Pile membership (no post-hoc split) is precisely the design discipline they call for.

Study	Outcome	Detectors	Statistical method	Conclusion
Shi et al. (2024; J. Zhang et al., 2025)	membership	reference-free (Min-K%/++)	AUC / TPR@FPR	detector raises membership AUC
Duan et al. (2024) (MIMIR)	membership	ref-free + reference	AUC on ground truth	MIAs ≈ chance on LLMs
Carlini et al. (2022) (LiRA)	membership	shadow/reference	TPR at low FPR	strong only with shadow models

B. Chen et al. (2025)	membership	reference-free	seed-variance testing vs loss	not robustly beyond loss
Hayes et al. (2025)	membership & extraction	LiRA (reference)	direct (zero-order) correlation	MIA does not imply extraction
Al Sahili et al. (2025)	extraction (targeted)	ref-free + AdaBoost	ranking precision, ensemble	marginal gains over likelihood
This work	extraction	ref-free calibrated	partial corr. + mediation (control loss)	no positive residual beyond loss

Where this work sits. To our knowledge it is the only study that pairs a per-item *extraction* outcome with a *partial-correlation/mediation* control for raw loss on *calibrated reference-free* detectors, yielding a quantified null-to-negative marginal that we read conservatively as no positive residual. Our row is measured on a single 160M model with $N = 300$ members and is preliminary (Section [3](#sec:limitations)).

Evaluation Overview

Threat model and success criteria

Section [3](#sec:threat) fixes the parties, the goals G1 through G3, the access tiers, and the success criteria. This section instantiates that protocol. Each detector is evaluated at its minimum access tier as graded in Section [3](#sec:threat), and success is defined by the low-false-positive operating point of Section [2.4](#sec:concepts) rather than by average accuracy.

Methods under comparison

We evaluate *existing* detectors only. We propose no new detector. The four detectors are defined in Section [2.4](#sec:concepts): LOSS/perplexity (Yeom et al., 2018), Min-K% Prob (Shi et al., 2024), Min-K%++ (J. Zhang et al., 2025), and the zlib-entropy ratio (Carlini et al., 2021). We run Min-K% and Min-K%++ at $k = 20$. Two further tests, also defined in Section [2.4](#sec:concepts), operate off the per-item likelihood axis: corpus-side n -gram overlap (Brown et al., 2020), which we use to construct ground-truth contamination labels for benchmark items, and the Oren permutation/exchangeability test (Oren et al., 2024) at the benchmark level. The leakage outcome is prefix-continuation extractable memorization under greedy decoding (Carlini et al., 2023). On the controlled corpus we additionally measure regex-detected PII leakage, framed via the PII-leakage games of Lukas et al. (2023). This measurement returned a null at our scale, and we report it as such rather than as a contribution (Section [6.3](#sec:res-extraction)). Related approaches we deliberately *do not* evaluate, guided prompting (Golchin & Surdeanu, 2024), neighbourhood and shadow-model reference attacks (Mattern et al., 2023; Shokri et al., 2017), and the divergence-style extraction of production models (Nasr et al., 2025), are discussed in Section [4](#sec:relatedwork). An internal-activation probe is reported, if at all, only as exploratory analysis in the Discussion, not as a contribution.

Data

Table [2](#tab:datasets) summarizes the corpora and benchmarks used or referenced below.

Models and corpus.

The primary model is the Pythia suite (Biderman et al., 2023), trained on the public Pile (Gao et al., 2020). Its reconstructible training order, 154 checkpoints, multiple sizes, and deduplicated variant provide exact membership ground truth. Our confound-clean split draws members from a public mirror of the Pile training set and non-members from the Pile validation set, stratified across the Pile’s constituent subsets. We additionally report the temporally confounded WikiMIA split as a deliberate contrast, and we cite the MIMIR construction (Duan et al., 2024) as the methodological precedent rather than using its released splits. OLMo (Groeneveld et al., 2024) on Dolma (Soldaini et al., 2024) is a secondary replication target. The Pile sits within the broader weakly filtered web-scrape regime, Common Crawl (Common Crawl Foundation, n.d.) and its filtered derivatives C4 (Dodge et al., 2021; Raffel et al., 2020) and RedPajama (Weber et al., 2024), that makes benchmark contamination structural rather than adversarial.

Dataset	Type	What it is	Size	Cite
The Pile	corpus	Curated 22-subset English corpus. Pythia’s training data and our membership ground truth	825 GB	(Gao et al., 2020)
pile-10k	corpus sample	Public 10k-document sample of the Pile training set, the source of our member items	10,000 documents	(Gao et al., 2020)
Enron Emails	corpus subset	Email corpus that is a component subset of The Pile. Supplies the PII measurement reported as a null	part of the Pile	(Gao et al., 2020)
Common Crawl	corpus	Open, continually updated repository of raw web-crawl data, the base of most LLM pre-training scrapes	petabyte-scale (growing)	(Common Crawl Foundation, n.d.)
C4	corpus	Colossal Clean Crawled Corpus: a filtered Common Crawl snapshot introduced with T5	~750 GB	(Dodge et al., 2021; Raffel et al., 2020)
Dolma	corpus	Open pre-training corpus, OLMo’s training data (replication target)	3 T tokens	(Soldaini et al., 2024)
RedPajama	corpus	Open reproduction of an LLaMA-style pre-training mixture	~30 T tokens	(Weber et al., 2024)

WikiMIA	benchmark	Membership-inference benchmark split by Wikipedia edit date. Used only as a deliberately temporally confounded contrast	64-token split	(Shi et al., 2024)
MMLU	benchmark	Multiple-choice knowledge/reasoning across 57 subjects	15,908 questions	(Hendrycks et al., 2021)
GSM8K	benchmark	Grade-school multi-step math word problems	8,500 problems	(Cobbe et al., 2021)
HumanEval	benchmark	Hand-written Python programming problems with unit tests	164 problems	(M. Chen et al., 2021)
HellaSwag	benchmark	Adversarially filtered commonsense sentence completion	~70,000 items	(Zellers et al., 2019)
TruthfulQA	benchmark	Questions probing imitative falsehoods	817 questions	(Lin et al., 2022)
BoolQ	benchmark	Naturally occurring yes/no reading-comprehension questions	15,942 questions	(Clark et al., 2019)

Corpora and benchmarks used in or referenced by the evaluation. The Pile is our ground-truth training corpus, and the Enron Emails subset supplies the PII measurement reported as a null in Section [6.3](#sec:res-extraction). WikiMIA appears only as the deliberately confounded contrast split. The lower block lists the contamination benchmarks, of which MMLU, GSM8K, and HumanEval are labeled by corpus-side overlap at the scale reported here.

Benchmarks and PII.

Contamination is tested against MMLU, GSM8K, and HumanEval at the scale reported here. HellaSwag, TruthfulQA, and BoolQ are configured in the harness and included in Table [2](#tab:datasets) for reference, and are left for the GPU replication. For the PII measurement we use the Enron Emails data *as a Pile subset already present in Pythia’s training data*, rather than fine-tuning a model to memorize PII. The measurement returned no detected leakage at this scale and is reported as a null (Section [6.3](#sec:res-extraction)). No real PII is reproduced in the paper.

Metrics

The rationale for reporting true-positive rate at a low fixed false-positive rate rather than average accuracy is given in Section [2.4](#sec:concepts). Following that convention (Carlini et al., 2022), the primary metric is *true-positive rate at a fixed low false-positive rate* (TPR @ 0.1% and 1% FPR) reported with *log-scale ROC*. AUC-ROC is reported secondarily. For benchmark flagging at a chosen operating threshold we additionally report precision/recall/F1 as a secondary, application-facing view. The leakage outcome is the *extraction rate* (Carlini et al., 2023). The headline analysis is the *Spearman correlation between per-item contamination score and per-item extraction outcome*, with bootstrap confidence intervals and a pre-registered partial-correlation control that isolates the contribution of raw loss, the quantitative form of RQ2.

Validation and controls

Robustness is established by bootstrap confidence intervals on TPR@FPR and on the Spearman correlation, and by a permutation/exchangeability test for benchmark-level contamination (Oren et al., 2024). We run the ablations that preempt the standard confounds and that are feasible at this scale: deduplicated versus non-deduplicated Pythia (duplication) and frequency controls (string frequency). The model-size scaling arm, which asks whether the detector-to-extraction relationship strengthens with scale as memorization does (Carlini et al., 2023), is built into the pipeline but not run here, and is reported as GPU-gated in Section [8](#sec:limitations). Differentially private training (Abadi et al., 2016; Li et al., 2022) is discussed as the mitigation direction (Section [4.5](#sec:dp)), not implemented, since it is a producer-side defense applied at training time rather than an auditor-side detector.

This section instantiates the protocol of Section [3](#sec:threat) over concrete methods, data, and metrics. The empirical results under this protocol, per-detector TPR at low FPR with log-scale ROC, extraction rates, and the headline detector-to-extraction correlation with confidence intervals, are reported in the results section, with every reported number tracing to a logged harness run.

Results

All results in this section are preliminary, obtained on Pythia-160M on CPU with $N = 300$ ground-truth Pile members (seed 0), and larger-model rows are left for the GPU replication. Every number is reproducible from a seeded script and recorded in our results ledger.

Membership separation is at chance on a confound-clean split

We first reproduce, as a control, the known weakness of membership inference on pre-trained LLMs (Duan et al., 2024). On a confound-clean split (members = Pile train, non-members = Pile validation, stratified across 22 Pile subsets to match domain), all four detectors sit at chance at 160M (Table [3](#tab:membership)). On the temporally-confounded WikiMIA split the same model shows a spurious 0.52–0.56, and a 1.4B model rises further, evidence that the WikiMIA signal is substantially distribution shift, not membership.

Construction (model)	LOSS	Min-K%	Min-K%++	zlib
Pile train-vs-val, clean (160M)	0.454	0.470	0.490	0.484
WikiMIA-64, confounded (160M)	0.523	0.539	0.545	0.564
WikiMIA-64, confounded (1.4B)	0.571	0.580	0.547	0.616

Membership AUC. Chance (≈ 0.5) on the confound-clean split at 160M. The WikiMIA “signal” is largely temporal/topical distribution shift. CIs in the ledger. Deduplicated Pythia gives the same chance-level result.

Detector scores predict extraction, but only through loss

Our headline analysis correlates each per-item detector score with the per-item extraction outcome (prefix-continuation extractable memorization under greedy decoding (Carlini et al., 2023)), then controls for raw loss. Table [4](#tab:headline) reports, for each calibrated detector, the zero-order Spearman ρ , the linear partial ρ given loss, the non-linear (cubic-residual) partial ρ with bootstrap CI, the FDR-corrected permutation q , and the mediation decomposition.

Detector	zero-order	partial loss	cubic-resid. [95 % CI]	BH- q	mediation: direct indirect
LOSS	+ 0.275	n/a	n/a	n/a	(mediator)
Min-K%	+ 0.173	− 0.178	− 0.110 [− 0.234, − 0.002]	0.058	− 0.394 + 0.567
Min-K%++	+ 0.108	− 0.148	− 0.160 [− 0.287, − 0.041]	0.015	− 0.213 + 0.321
zlib	+ 0.177	− 0.042	− 0.052 [− 0.165, + 0.068]	0.331	− 0.061 + 0.238

Headline: per-item contamination score vs. extraction (Spearman ρ), Pythia-160M, $N = 300$ members. The positive zero-order correlations collapse to ≈ 0 or significantly *negative* once loss is controlled, linearly, and under the non-linear cubic-residual control (no positive signal revives, and deciles and the deduplicated arm agree). Mediation: the loss-mediated *indirect* effect is significantly positive for all three detectors while the *direct* effect is null (zlib) or negative (Min-K%, Min-K%++). We read this as a *descriptive* decomposition, not a causal mediation claim (see below): no calibrated detector adds positive signal beyond loss.

Collinearity caveat (why we do not over-read the negative partials).

The calibrated detectors are deterministic transforms of the same per-token log-probabilities as loss, and are empirically collinear with it: Spearman $\rho(\text{loss}, \cdot) = 0.90$ (Min-K%), 0.74 (Min-K%++), 0.74 (zlib), with variance-inflation factors 6.2, 2.6, 2.4. Under the *linear* partial the strongest negative belongs to Min-K%, the most loss-collinear detector at VIF 6.2, which is the pattern a *suppression artifact* of near-collinearity would produce. That attribution does not survive the pre-registered primary control. Under cubic residualization the largest negative, and the only one significant after FDR correction, is Min-K%++ at VIF 2.6 (− 0.160, BH- $q = 0.015$), while Min-K% weakens to − 0.110 with $q = 0.058$ and a confidence interval whose upper bound is − 0.002. Because the surviving negative sits with a detector of only moderate collinearity, suppression alone does not account for it, and the attribution of these negatives is therefore unstable across near-equivalent specifications. We consequently do not build on them in either direction. We do not claim the calibrated detectors *negatively* predict extraction, and the defensible statement we carry forward is that they carry *no positive* extraction signal independent of loss.

The pre-registered decision rule asked whether any calibrated detector predicts extraction *beyond* loss (a positive partial ρ , CI excluding zero, FDR-significant). None does, under the linear or the non-linear control. **Power note:** with $N = 300$ and a near-degenerate outcome (3/300 fully extracted), this is evidence of *no positive independent signal of appreciable size*, not proof of an exact null. The analysis is well-powered only for moderate-to-large positive residuals, and a small positive effect at scale is not excluded (hence the GPU replication). The per-domain breakdown (ledger) shows the loss↔extraction link is heterogeneous and sign-flipping across domains, strongest in templated/structured domains (GitHub, StackExchange), reversed in some prose domains (PubMed Abstracts), so the pooled ρ is a domain-mixture, not a uniform effect.

Extraction and PII at this scale

Extractable memorization is rare at 160M: 3/300 members are fully extractable (exact-match extraction rate 0.010, mean fractional extraction 0.037), the fully-extracted items being templated boilerplate. On the Enron-Emails-in-Pile subset we measured *zero* verbatim PII leakage (8/36 documents contained PII in the held suffix, none were regurgitated). We report the PII result as a null at this scale and make no PII-exposure claim. Both quantities are expected to grow with model scale.

Benchmark contamination (model-free *n*-gram + permutation test)

We complement the per-item analysis with two benchmark-level contamination tests (Table [5](#tab:matrix), data-reference-type="ref", data-reference="tab:matrix">5). The model-free *n*-gram overlap against a public *sample* of the Pile (10k documents) is a scale-invariant method but, with a sampled reference, yields only a loose *lower bound*: overlap is near-zero for MMLU (0.2% at 13-grams), GSM8K (0%), and HumanEval (0% at 13-grams), which certifies overlap is *at least* this small and is uninformative about true contamination, a full-Pile index (infrastructure-, not GPU-, gated) is required for a real rate. The Oren permutation/exchangeability test (Oren et al., 2024) at 160M finds the canonical ordering favoured beyond chance for MMLU ($p = 0.001$) and GSM8K ($p = 0.013$) but not HumanEval ($p = 0.875$). We draw *no* contamination conclusion from this, as the test is membership-based, run at sanity scale (small k , smallest model), and subject to a fluency/orientation artifact, it is flagged GPU-gated and requires a fluency-control baseline before any claim.

Benchmark	13-gram overlap (lower bound)	8-gram overlap	Oren p (160M, sanity)
MMLU	0.2%	0.8%	0.001
GSM8K	0.0%	0.0%	0.013
HumanEval	0.0%	1.8%	0.875

Benchmark-level contamination at small scale. *n*-gram cells are a *lower bound* against a 10k Pile sample (method scale-invariant, reference under-powered). Oren p -values are sanity-scale at 160M and GPU-gated (no contamination conclusion drawn). See docs/contamination_matrix.md.

Discussion

The detectors carry little leakage information that loss does not.

Hayes et al. (2025) already report that membership-attack success and extraction come apart. Our contribution is the measurement-validity question that follows: the contamination signal which predicts *extraction* is, to the resolution of our experiment, *just raw loss*. The reference-free detectors that the contamination-detection literature has invested in, Min-K%, Min-K%++, zlib, improve membership ranking by re-calibrating the per-token likelihood (z-scoring against the vocabulary, compressing, or trimming to the lowest-probability tokens), and one reading consistent with our data is that these adjustments remove loss-magnitude information that tracks extractability. We cannot separate that reading from suppression under near-collinearity, so we do not assert it as a mechanism. A descriptive mediation decomposition is consistent with this, the loss-mediated (indirect) path is positive for all three detectors while the direct paths are null or negative, but we read it descriptively, not causally: the detectors are near-collinear transforms of loss (Spearman up to 0.90, VIF up to 6.2), so a

negative direct/partial term is consistent with statistical suppression rather than genuine inverse prediction. We therefore claim only the conservative version: the calibrated detectors add *no positive* leakage signal beyond loss. A practitioner who wants to know *which contaminated items the model will actually leak* is, on this evidence, no better served by a state-of-the-art membership detector than by raw loss. This gives a mechanistic form to the divergence that Hayes et al. (2025) first reported, on the reference-free detectors an auditor can actually afford.

Why this is a security result, not a leaderboard result.

Our finding is deliberately *not* “we built a better detector.” It is that the privacy question, will contamination of a benchmark expose a leakage channel? is mis-served by importing the membership-inference toolkit wholesale. For an auditor of a released model, the actionable implication is to measure loss/extractability directly and to treat a high Min-K%/Min-K%++ score as evidence about membership, not about leakage risk. The contribution is the assumption chain of Section [2.5](#sec:chain) plus this controlled measurement of incremental value. The detectors themselves are prior work. Every statement in this section is bounded by the regime it was measured in: one 160M model, $N = 300$ members, three fully extracted items, and membership AUC at chance. It is a bound on what these instruments demonstrably carry, not a demonstration that they carry nothing at scale.

Relation to prior work.

Our direction agrees with published results and we do not claim the bottom line is surprising. Hayes et al. (2025) established the divergence between membership success and extraction, and Al Sahili et al. (2025) report only “marginal” gains of MIA scores over likelihood ranking for targeted extraction. What we add is the controlled form of the claim, a pre-registered partial-correlation/mediation that quantifies a *zero-to-negative* residual for the calibrated reference-free detectors after loss is removed, and we target the reference-free detectors the contamination literature actually deploys rather than a shadow-model attack. B. Chen et al. (2025) independently find these detectors do not robustly beat the loss baseline for *membership* once seed variance is accounted for. Our result is the extraction-outcome analogue.

Defenses.

Because the leakage we measure is downstream of memorization, the principled mitigation is differential privacy applied at training time (Abadi et al., 2016; Li et al., 2022). It is a producer-side control, not an auditor-side detector, and bounds the very quantity (loss-magnitude / memorization) our analysis identifies as the operative one.

Limitations

We state the limitations plainly. Several bound the strength of the present claims and motivate the GPU-scale replication the pipeline is built for.

- **A3 and A4 are untested.** Our design bears on A1 and secondarily on A2 (Section [2.5](#sec:chain)). Testing A3 would require a corpus with known sensitive content and a model scale at which extraction is not near-degenerate. Testing A4 would require measuring the same detector-to-extraction relationship separately on sensitive records and on benchmark-like documents, since memorization is example-dependent and duplication-sensitive. Neither is possible at the scale reported here.

- **Single, smallest model.** All results are on Pythia-160M (CPU). Memorization grows log-linearly with model scale (Carlini et al., 2023), so both the membership signal and the extraction outcome are expected to be stronger at 1.4B–12B. The present numbers are *preliminary*. We have built every analysis so the larger-model run is a one-line configuration change.
- **Chance-level membership separation.** On the confound-clean Pile train-vs-val split, membership AUC is at chance (0.45–0.49) at 160M, consistent with (Duan et al., 2024). Our incremental-value result is therefore established in a regime where the membership signal is itself weak. Whether the calibrated detectors gain *independent* extraction-predictive value once membership separation becomes non-trivial at scale is an open question our design is poised to answer.
- **Near-degenerate extraction outcome.** Extractable memorization at 160M is rare (3/300 items fully extracted, mean fractional extraction 0.037), so the correlation analysis leans on a small high-extraction tail. We mitigate with rank statistics, bootstrap CIs, and a zero-robust Kendall check, but a less zero-inflated outcome at scale would sharpen all estimates.
- **PII leakage is a reported null, not a pending result.** On the Enron-in-Pile subset we observed *zero* verbatim PII leakage at 160M (8/36 documents contained PII in the held suffix, none were regurgitated). We report this as a null and make no PII-exposure claim. It is the reason assumption A3 of Section [data-reference-type="ref" data-bbox="220 428 240 441" style="color: #000080;">data-reference="sec:chain">2.5](#sec:chain) is untested here rather than supported.
- **Benchmark-level test underpowered.** The Oren permutation/exchangeability test is run only at sanity scale on 160M. Membership-based, it is underpowered here and is flagged as GPU-gated rather than used to draw contamination conclusions.
- ***n*-gram contamination is a lower bound.** Our model-free *n*-gram overlap uses a public *sample* of the Pile as the reference index, so measured benchmark↔Pile overlap underestimates the true overlap against the full corpus.
- **Observational, members-only correlation.** The headline analysis correlates detector scores with extraction across known members. It is observational, not interventional. We address the most important confound (loss) by pre-registered partial correlation and mediation, and the obvious alternatives (frequency, duplication, non-linearity, distribution shift) by explicit controls, but residual confounding cannot be excluded.
- **Collinearity of detectors with loss.** The calibrated detectors are deterministic transforms of the same per-token log-probabilities as loss and are empirically collinear with it (Spearman 0.74–0.90, VIF up to 6.2 for Min-K%). Consequently we interpret the negative partial/direct terms as possible *suppression artifacts* of near-collinearity and claim only the conservative “no positive residual” result. We do not assert the detectors inversely predict leakage.
- **Construct validity of the extraction proxy.** The outcome (greedy prefix-continuation extraction over the held suffix) is itself likelihood-related, so part of the loss↔ extraction association is mechanical/definitional. Our control removes the loss component, but a decisive separation would compute prefix-only loss against extraction. We flag this as a known construct-validity limitation rather than claiming the two are independent by construction.

- **Selection and aggregation.** Members are drawn from a non-uniform public Pile sample (pile-10k), so member-selection bias is possible, and the pooled correlation aggregates domains whose effects flip sign (Section [6.2](#sec:res-headline)), so the pooled ρ should be read as a domain-mixture, not a homogeneous effect.
- **Linearity (now addressed).** An earlier version controlled for loss only linearly. We added a cubic-residual and decile-stratified non-linear control, under which no positive independent signal revives. We note it here because it was a live threat to the claim until tested.

Conclusion

We took the common argument that benchmark contamination is a privacy warning sign, restated it as four explicit assumptions, and tested the first of them on models whose training corpus is public. The question was whether the contamination signal an auditor can cheaply compute predicts concrete extraction once the model's per-item loss is held fixed. Using a pre-registered partial-correlation and mediation analysis that controls for raw per-item loss, we found that it does, but only through loss: the calibrated reference-free detectors (Min-K%, Min-K%++, zlib) add no positive independent predictive value beyond loss. The two negative residuals we observe are consistent with statistical suppression under near-collinearity rather than with inverse prediction, and we claim only the absence of positive value. The result is robust to a non-linear loss control and to deduplication, and is not a frequency or zero-inflation artifact. The practical message is a measurement-validity bound: the detectors optimized for membership inference carry little information about *which* items leak that loss does not already carry, so an auditor should measure loss and extractability directly. We claim no new detector or metric. The contributions are the explicit assumption chain of Section [2.5](#sec:chain) and the controlled, pre-registered measurement of incremental value. These findings are preliminary, on the smallest Pythia model. The immediate next step, and the design target of our released pipeline, is the GPU-scale replication across model sizes, where memorization and extraction are expected to strengthen, and where assumptions A3 and A4 of Section [2.5](#sec:chain) could be tested rather than assumed, and where the question of whether calibrated detectors gain independent leakage-predictive value at scale can be settled.

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